

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

3. PROJECT DESIGN PHASE

3.1 Problem-Solution Fit

The Student Enrollment & Course Management System addresses the challenges of manual student registration, course enrollment, and academic record management.

The proposed system provides a centralized digital platform to simplify enrollment activities, maintain accurate student records, and improve course management.

3.2 Proposed Solution

Aspect	Design
Problem	Manual student enrollment and course management can cause data-entry errors, duplicate records, data loss, and delays in processing enrollment information.
Solution	A web-based Student Enrollment & Course Management System with centralized student records, course information, and secure user access.
Innovation	Digital student registration, automated enrollment processing, data validation, centralized storage, and real-time enrollment status tracking.
Impact	Improved data accuracy, faster enrollment processing, reduced paperwork, and better academic administration.
Business Model	A digital academic management platform for educational institutions to manage student enrollment and course-related activities.
Scalability	Can support increasing numbers of students, courses, faculty members, and educational institutions as the system grows.

3.3 Four-Layer Architecture

Presentation Layer (Frontend)

- Provides a user-friendly interface for students, faculty, and administrators.
- Supports registration, login, course viewing, and enrollment activities.

2. Application Layer (Backend)

- Handles student registration, course enrollment, authentication, and business logic.
- Processes user requests and manages system operations.

3. Data Layer (Database)

- Stores student details, course information, enrollment records, and user accounts.
- Maintains accurate and centralized data.

4. Security Layer

- Provides secure login, role-based access control, and data protection.
- Ensures users can access only authorized features and information.

3.4 Object and Relationship Design

- Student Object: Stores student ID, name, email, contact number, and address.
- Course Object: Stores course ID, course name, description, duration, and fees.
- Enrollment Object: Stores enrollment ID, student ID, course ID, enrollment date, and status.
- Faculty Object: Stores faculty ID, name, email, contact details, and assigned courses.
- User Object: Stores login credentials, user roles, and access permissions.

3.5 Design Goals

The main design goal of the Student Enrollment & Course Management System is to develop a user-friendly, secure, and efficient platform for managing student registration, course enrollment, and academic records. The system aims to simplify enrollment activities, maintain accurate and centralized student information, reduce manual paperwork, and minimize data-entry errors. It provides easy access to course details and enrollment status while supporting role-based access for students, faculty, and administrators. The system is designed to ensure data security, faster processing, reliable performance, easy maintenance, and scalability to support the growing needs of educational institutions.